FINAL REPORT DOCUMENT

FOR

KCAU SOCIAL MEDIA APPLICATION

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CHAPTER ONE: INTRODUCTION

1.0 Background

The rapid growth of digital communication technologies has significantly transformed how individuals interact, share information, and collaborate. Social media platforms have become essential tools for communication, learning, and knowledge sharing across various communities, including academic institutions.

At **KCA University,** students rely on various fragmented communication channels such as messaging applications, emails, and informal social media groups to exchange information and share academic resources. Although these platforms provide basic communication functionalities, they are not specifically designed to address the academic and social needs of university students in a centralized and structured manner.

The **KCAU Social Media Application** was developed to address this gap by providing a centralized platform tailored specifically for KCA University students. The system allows users to create accounts, share posts, upload academic resources, interact through comments, and connect with other students within the institution. By integrating social interaction with academic resource sharing, the platform aims to enhance collaboration, improve communication efficiency, and facilitate easier access to information among students.

2.0 Problem Statement

Students at KCA University currently use multiple disconnected platforms such as messaging apps, emails, and social media groups to communicate and share academic resources. These platforms are not centralized or specifically designed for academic collaboration, leading to scattered information, difficulty in accessing study materials, and limited interaction among students.

As a result, there is no single, organized system where students can efficiently communicate, share resources, and collaborate. This creates inefficiencies in information sharing and reduces opportunities for academic engagement.

Therefore, there is a need for a centralized platform that integrates communication and resource sharing for KCA University students.

3.0 Objectives

3.1 General Objective

To design and develop a centralized social media application that enhances communication and resource sharing among KCA University students.

3.2 Specific Objectives

- To design a user-friendly interface for student interaction
- To implement user registration, login, and profile management features
- To enable users to create, edit, and delete posts
- To allow students to upload and share academic resources
- To implement a commenting and interaction system for posts
- To ensure secure authentication and data handling within the system
- To test and evaluate the system for usability and performance

4.0 Significance of the Project

The KCAU Social Media Application is expected to provide several benefits to students and the university as a whole:

- **Enhanced Communication:**
  The platform provides a centralized space where students can easily communicate and interact.
- **Improved Resource Sharing:**
  Students can upload and access academic materials conveniently, improving learning efficiency.

- **Increased Collaboration:**
  The system encourages collaboration among students across different courses and academic levels.
- **Time Efficiency:**
  Reduces the time spent searching for information across multiple platforms.
- **Academic Support:**
  Facilitates peer-to-peer learning and knowledge exchange.
- **Foundation for Future Development:**
  The system can be expanded to include additional features such as notifications, group discussions, and integration with university systems.

CHAPTER TWO: LITERATURE REVIEW AND SYSTEM REQUIREMENTS

2.1 Literature Review

Social media platforms such as WhatsApp, Telegram, and Facebook are widely used by students for communication and sharing academic materials. In university settings, these platforms are often used to exchange notes, discuss coursework, and share resources such as past papers.

However, these platforms are not designed for structured academic use. Academic materials shared in group chats are often disorganized and difficult to retrieve. For example, past examination papers and important documents are frequently buried in long message threads, making them hard to find when needed. New students may also struggle to access previously shared resources since there is no centralized or searchable repository.

This lack of organization leads to inefficiencies in accessing academic materials and limits effective collaboration among students. Research shows that centralized systems designed for academic purposes improve accessibility, organization, and sharing of learning resources.

Therefore, a dedicated social media platform tailored for students can address these challenges by providing a structured environment where academic resources such as past papers can be easily stored, organized, and accessed.

2.2 System Requirements Specification (SRS)

This section outlines the requirements of the KCAU Social Media Application, including functional and non-functional requirements.

2.2.1 Functional Requirements

These define what the system should do:

- The system shall allow users to register and create accounts
- The system shall allow users to log in and log out securely
- The system shall allow users to create, edit, and delete posts
- The system shall allow users to upload and share academic resources
- The system shall allow users to view posts from other users
- The system shall allow users to comment on posts
- The system shall allow users to manage their profiles (e.g., bio, profile picture)

2.2.2 Non-Functional Requirements

These define how the system should perform:

- **Usability:** The system should be user-friendly and easy to navigate
- **Performance:** The system should respond quickly to user actions
- **Security:** User data should be protected through authentication and authorization
- **Scalability:** The system should support an increasing number of users
- **Reliability:** The system should operate consistently with minimal downtime

2.2.3 System Users

- **Students:** Main users who can register, post, share resources, and interact
- **Administrator:** Manage users and monitor system activity

CHAPTER THREE: METHODOLOGY

3.1 Data Collection Methods

Data for the development of the KCAU Social Media Application was collected using the following methods:

- **Questionnaires:**
  Questionnaires were distributed to students to gather information on how they currently communicate and share academic resources, and to identify the challenges they face (e.g., difficulty accessing past papers).
- **Interviews:**
  Informal interviews were conducted with students to gain deeper insights into their needs and expectations for a centralized platform.
- **Observation:**
  Existing student communication platforms such as WhatsApp groups were observed to understand how students share notes, past papers, and other resources.
- **Document Review:**
  Relevant academic materials and system requirements documents were reviewed to guide the system design.

3.2 System Design Specifications

The system was designed to provide a centralized and user-friendly platform for communication and resource sharing.

- **Architecture Design:**
  The system follows a client-server architecture where users interact with

the application through a web interface, and data is processed and stored on the server.

- **User Interface Design:**
  The interface was designed to be simple and intuitive, allowing users to easily navigate through features such as posting, commenting, and uploading resources.
- **Database Design:**
  A relational database was used to store user data, posts, comments, and uploaded resources.
- **Key Modules:**
  - User Authentication Module (registration and login)
  - Post Management Module (create, edit, delete posts)
  - Resource Sharing Module (upload and access materials like past papers)
  - Interaction Module (comments and engagement)

3.3 Test Plan

The system was tested to ensure it meets the specified requirements and functions correctly.

- **Unit Testing:**
  Individual components such as login, posting, and file upload were tested separately.
- **Integration Testing:**
  Different modules were tested together to ensure they work as a complete system.
- **System Testing:**
  The entire application was tested to verify that all features function as expected.
- **User Acceptance Testing (UAT):**
  Selected students tested the system and provided feedback on usability and functionality.
- **Test Cases Included:**
  - User registration and login
  - Creating and deleting posts
  - Uploading and accessing resources
  - Commenting on posts

3.4 Implementation Plan

The system was implemented in phases to ensure smooth development and deployment.

- **Phase 1: Planning and Requirement Analysis**
  Identification of system requirements and user needs.
- **Phase 2: System Design**
  Designing the system architecture, database, and user interface.
- **Phase 3: Development**
  Coding of system features including authentication, posting, and resource sharing.
- **Phase 4: Testing**
  Conducting various tests to identify and fix errors.
- **Phase 5: Deployment**
  Making the system available for use by students.
- **Phase 6: Maintenance**
  Monitoring the system and making improvements based on user feedback.

CHAPTER FOUR: SYSTEM TESTING AND RESULTS

4.1 Graphical User Interface Test Results

The Graphical User Interface (GUI) of the KCAU Social Media Application was tested to ensure usability, responsiveness, and correctness of user interactions.

Test Case	Description	Expected Result	Actual Result	Status
Login Page	User enters valid credentials	User is redirected to dashboard	Successfully redirected	Pass
Login Page	User enters invalid credentials	Error message displayed	Error message shown	Pass
Registration Page	User fills all required fields	Account created successfully	Account created	Pass
Home Feed	User views posts	Posts displayed correctly	Posts displayed	Pass
Create Post	User submits a post	Post appears on feed	Post displayed	Pass
Comment Feature	User comments on a post	Comment is added	Comment displayed	Pass
Resource Upload	User uploads file (e.g., past paper)	File uploaded successfully	File uploaded	Pass

The GUI was found to be user-friendly, responsive, and functional. All tested features performed as expected without major usability issues.

4.2 Database Test Cases

Database testing was conducted to ensure correct storage, retrieval, and integrity of data.

Test Case	Description	Expected Result	Actual Result	Status
User Registration	Save new user data	Data stored in database	Data saved correctly	Pass
Login	Retrieve user	Correct user	User	Pass

Authentication	credentials	validated	authenticated	
Create Post	Save post data	Post stored in database	Post saved	Pass
Comment Storage	Save comments	Comment linked to post	Comment saved correctly	Pass
File Upload	Store file path/reference	File accessible from database	File retrieved successfully	Pass
Delete Post	Remove post from database	Post deleted permanently	Post removed	Pass

4.3 System Output Test Cases

System output testing was carried out to verify that the system produces correct and expected results based on user actions.

Test Case	**Input**	**Expected Result**	**Actual Result**	**Status**
Valid Login	Correct username & password	User dashboard displayed	Dashboard shown	Pass
Invalid Login	Wrong credentials	Error message	Error displayed	Pass
Post Creation	User submits text	Post appears on feed	Post shown	Pass
Resource Access	User clicks uploaded file	File opens/downloads	File accessed	Pass
Comment Submission	User adds comment	Comment appears under post	Comment displayed	Pass

CHAPTER FIVE: CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

The KCAU Social Media Application was successfully designed and developed to provide a centralized platform for communication and academic resource sharing among students at KCA University. The system addressed the problem of fragmented communication channels by integrating key features such as user authentication, post creation, commenting, and resource sharing into a single platform.

The system was tested and confirmed to function correctly, with all major features operating as expected. Users are able to easily share and access academic materials such as notes and past papers, improving collaboration and access to information.

Overall, the project met its objectives by delivering a functional, user-friendly, and efficient system that enhances student interaction and academic engagement.

5.2 Contributions

This project made several important contributions:

- **Centralized Platform:**
  Provided a single system where students can communicate and share academic resources efficiently.
- **Improved Resource Accessibility:**
  Made it easier for students to access important materials such as past papers without searching through multiple platforms.
- **Enhanced Collaboration:**
  Encouraged interaction and knowledge sharing among students across different courses.

- **Practical Application of Skills:**
  Demonstrated the use of web development technologies (e.g., Django, database systems) in solving real-world problems.

- **Foundation for Future Systems:**
  Established a base system that can be expanded with additional features in the future.

5.3 Recommendations

Although the system was successfully developed, several improvements can be made in future work:

- **Mobile Application Development:**
  Develop a mobile version of the system to improve accessibility and user experience.
- **Notification System:**
  Add real-time notifications for posts, comments, and new resources.
- **User Groups and Forums:**
  Introduce group discussions based on courses or units.
- **Integration with University Systems:**
  Integrate the platform with existing university systems for better information sharing.
- **Enhanced Security Features:**
  Improve security through features such as email verification and stronger authentication mechanisms.

REFERENCES

Django Software Foundation. (n.d.). *Django Documentation*. Retrieved from https://docs.djangoproject.com

Pressman, R. S. (2014). *Software Engineering: A Practitioner's Approach* (8th ed.). McGraw-Hill Education

Sommerville, I. (2016). *Software Engineering* (10th ed.). Pearson

APPENDICES

Appendix I – III: Work Progress Documents

These appendices contain documents developed during the project lifecycle. They provide evidence of the planning, analysis, and design stages of the KCAU Social Media Application.

Appendix I: Project Proposal

Introduction:
The KCAU Social Media Application is a centralized platform designed to improve communication and resource sharing among students at KCA University. It allows students to create accounts, share posts, upload academic resources, and interact with peers.

Problem Statement:
Students currently rely on fragmented communication channels such as WhatsApp, emails, and informal social media groups. This makes it difficult to share and access academic resources like past papers, resulting in inefficiencies and reduced collaboration.

Objectives:

- To develop a centralized platform for student communication and resource sharing
- To enable posting, commenting, and uploading of academic materials
- To ensure secure and user-friendly access to the system

Scope:
The system focuses on providing:

- Student registration and authentication
- Post creation and commenting
- Uploading and accessing academic resources
- A simple, intuitive interface for web use

Appendix II: Requirements Analysis

Functional Requirements:

- User registration and login/logout
- Post creation, editing, and deletion
- Resource upload and access (e.g., past papers)
- Commenting on posts
- Profile management

Non-Functional Requirements:

- Usability: Easy navigation and user-friendly
- Performance: Fast response to user actions
- Security: Data protection and authentication
- Reliability: Consistent operation with minimal downtime
- Scalability: Supports growing number of users

User Roles:

- Students: Can post, comment, upload resources
- Administrator (optional): Monitors system activity, manages users

Use Case Descriptions:

- **Register User:** Student provides information → Account created → User logs in
- **Create Post:** Student writes post → Post saved → Appears on feed
- **Upload Resource:** Student selects file → Upload → Accessible by other students

Appendix III: System Design

The system design of the KCAU Social Media Application focused on creating a centralized, user-friendly platform for student communication and academic resource sharing. The design consisted of the following components:

1. Architecture Design:
The system follows a client-server architecture. Users interact with the application through a web interface, while all data processing, storage, and

business logic are handled on the server. This ensures secure data handling and centralized management.

2. Database Design:
A relational database was designed to store all user information, posts, comments, and uploaded academic resources. Key tables include:

- **Users:** Stores student details, login credentials, and profile information
- **Posts:** Stores content shared by students along with timestamps and associated user IDs
- **Comments:** Stores comments linked to posts and users
- **Resources:** Stores uploaded files (e.g., past papers) and links them to the uploading user

3. User Interface Design:
The interface was designed to be intuitive and simple. Students can easily navigate to create posts, comment, upload resources, and access materials shared by others. Attention was given to layout, readability, and ease of access.

4. Module Design:
The system was divided into distinct functional modules to improve organization and maintainability:

- **User Authentication Module:** Handles registration, login, and profile management
- **Post Management Module:** Allows creation, editing, and deletion of posts
- **Resource Sharing Module:** Enables uploading and accessing academic materials
- **Interaction Module:** Facilitates commenting and engagement on posts

5. Security Design:
Measures were incorporated to ensure data security, including password encryption, controlled access to resources, and user authentication mechanisms.

6. Workflow Design:
The system workflow ensures smooth interaction for students:

- Students register and log in → Access the home feed → Create posts or upload resources → Interact with other students via comments → Log out

Appendix IV: User Training Manual

This section provides guidance on how to use the system.

1. User Registration

- Navigate to the registration page
- Enter required details (username, email, password)
- Click “Register”

2. User Login

- Enter username and password
- Click “Login” to access the system

3. Creating a Post

- Navigate to the home page
- Enter post content
- Click “Post”

4. Uploading Resources (e.g., Past Papers)

- Select upload option
- Choose file from device
- Click “Upload”

5. Commenting

- Select a post
- Enter comment
- Submit

Appendix V: Sample System Code

This appendix provides selected portions of the KCAU Social Media Application code to demonstrate the implementation of key functionalities. Only essential code snippets are included for clarity.

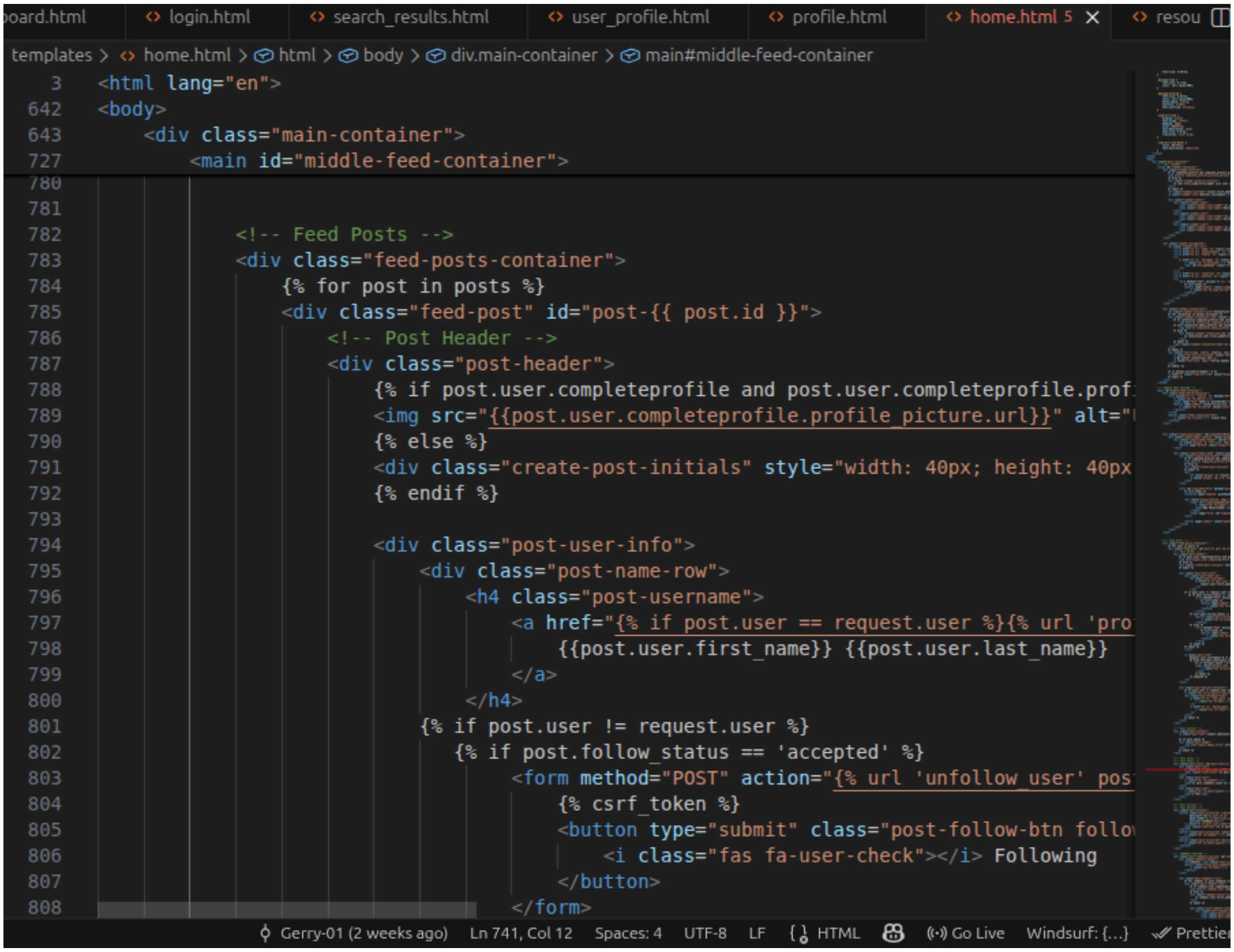
1. Django Model for Creating tables in the database

```python
from django.db import models
from django.contrib.auth.models import User
from django.utils import timezone

class Profile(models.Model):
    user = models.OneToOneField(User, on_delete=models.CASCADE)
    id_number = models.CharField(max_length=20, unique=True)
    def __str__(self):
        return self.user.username


class CompleteProfile(models.Model):
    user = models.OneToOneField(User, on_delete=models.CASCADE)
    CAMPUS_CHOICES = [
        ('main', 'Main Campus'),
        ('town', 'Town Campus'),
        ('kitengela', 'Kitengela Campus'),
        ('western', 'Western Campus'),
    ]
    campus = models.CharField(max_length=20, choices=CAMPUS_CHOICES)
    FACULTY_CHOICES = [
        ('computing', 'Faculty of Computing & Information Management'),
        ('business', 'School of Business'),
        ('education', 'School of Education'),
        ('arts', 'School of Arts & Social Sciences'),

    ]
    faculty = models.CharField(max_length=50, choices=FACULTY_CHOICES)
    PROGRAM_CHOICES = [
    ('bsd', 'Bachelor of Science in Data Science (BSD)'),
    ('bbit', 'Bachelor of Business Information Technology (BBIT)'),
    ('bisf', 'Bachelor of Information Security & Forensics (BISF)'),
    ('bac', 'Bachelor of Applied Communication (BAC)'),
```

2. Django View to connect front end and Backend

```python
from django.shortcuts import render, redirect, get_object_or_404
from django.contrib import messages
from django.contrib.auth.models import User
from django.contrib.auth import authenticate, login as auth_login, logout as auth_logout
from django.contrib.auth.decorators import login_required
from .models import Profile, CompleteProfile, Post, Comments, Resource, Event, AdminProfil
import random
import calendar
from datetime import datetime
from django.core.paginator import Paginator
from django.utils import timezone
from django.http import JsonResponse
from django.views.decorators.http import require_POST
from django.db.models import Q, Max



@login_required
def home(request):
    # Handle regular user profile (may not exist for admins)
    try:
        user_profile = Profile.objects.get(user=request.user)
    except Profile.DoesNotExist:
        user_profile = None

    try:
        complete_profile = CompleteProfile.objects.get(user=request.user)
    except CompleteProfile.DoesNotExist:
        complete_profile = None

    posts = Post.objects.all().order_by('-created_at')

```

3. Django Template to Display Posts

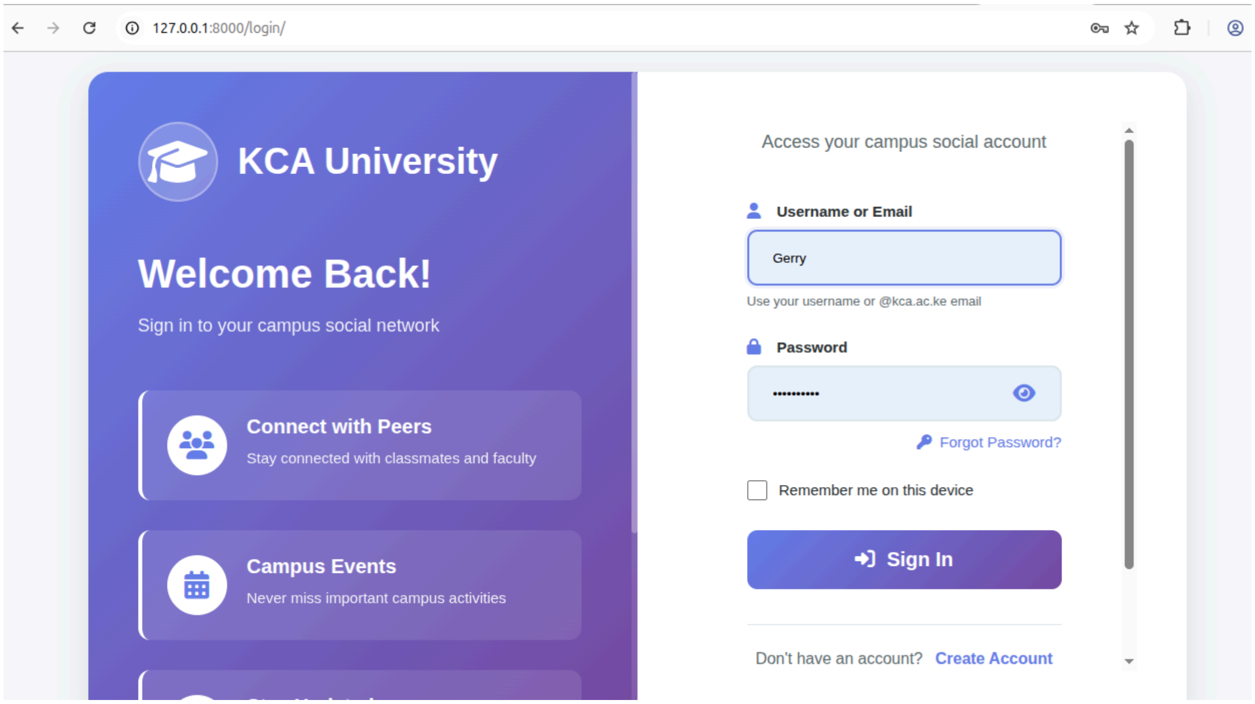


```
<html lang="en">
<body>
    <div class="main-container">
        <main id="middle-feed-container">


            <!-- Feed Posts -->
            <div class="feed-posts-container">
                {% for post in posts %}
                <div class="feed-post" id="post-{{ post.id }}">
                    <!-- Post Header -->
                    <div class="post-header">
                        {% if post.user.completeprofile and post.user.completeprofile.prof
                        <img src="{{post.user.completeprofile.profile_picture.url}}" alt="
                        {% else %}
                        <div class="create-post-initials" style="width: 40px; height: 40px
                        {% endif %}

                        <div class="post-user-info">
                            <div class="post-name-row">
                                <h4 class="post-username">
                                    <a href="{% if post.user == request.user %}{% url 'pro
                                        {{post.user.first_name}} {{post.user.last_name}}
                                    </a>
                                </h4>
                            {% if post.user != request.user %}
                               {% if post.follow_status == 'accepted' %}
                                    <form method="POST" action="{% url 'unfollow_user' pos
                                        {% csrf_token %}
                                        <button type="submit" class="post-follow-btn follo
                                            <i class="fas fa-user-check"></i> Following
                                        </button>
                                    </form>
```

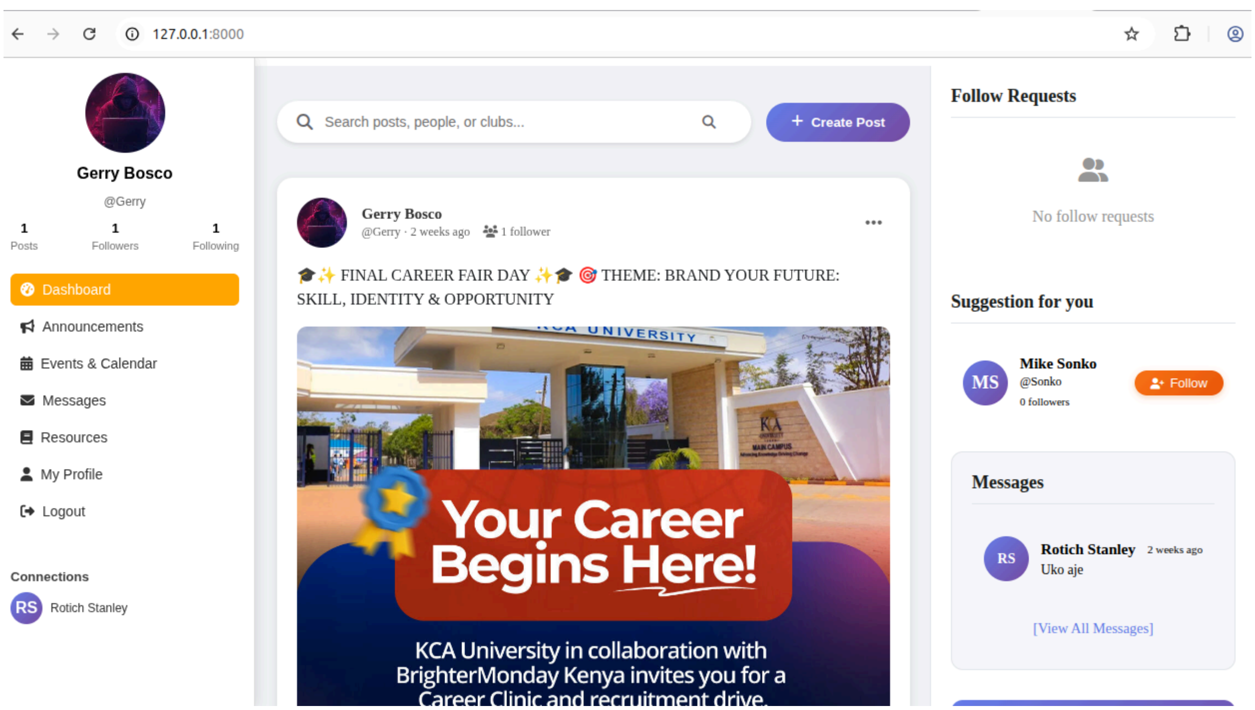
Appendix VI: Sample

This appendix contains sample screenshots of the KCAU Social Media Application to illustrate its key functionalities and user interface.

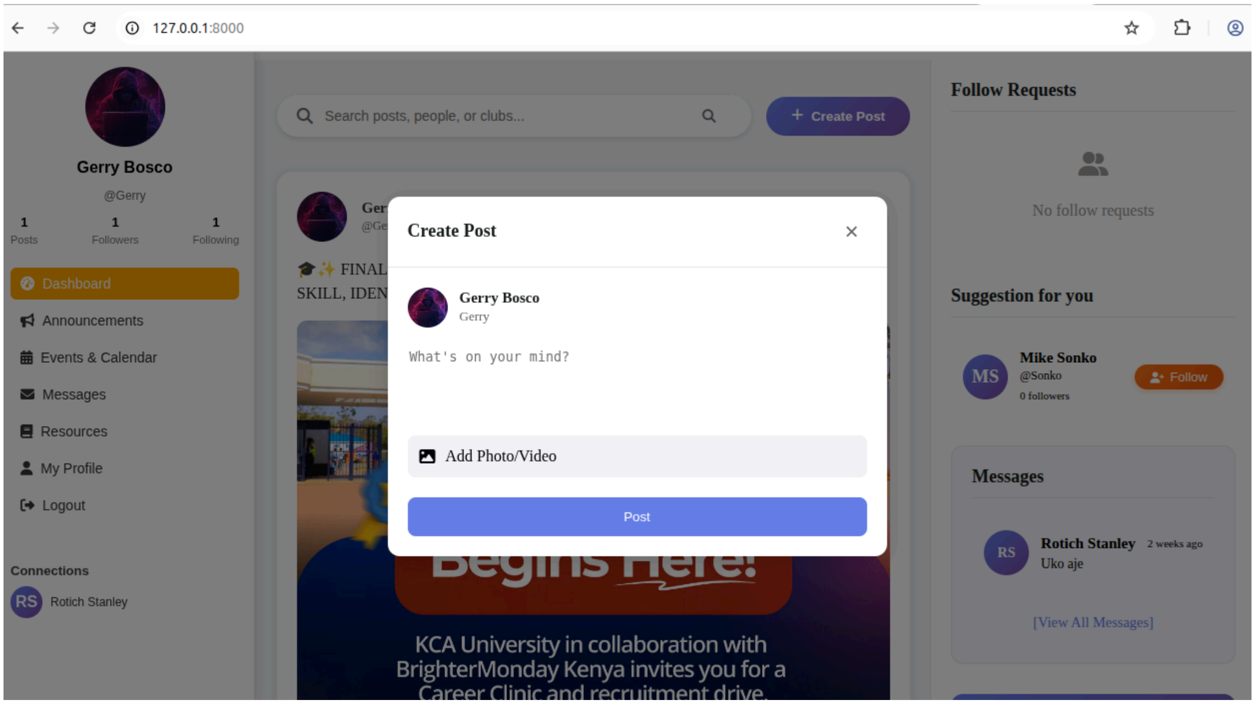
1. **Login page**



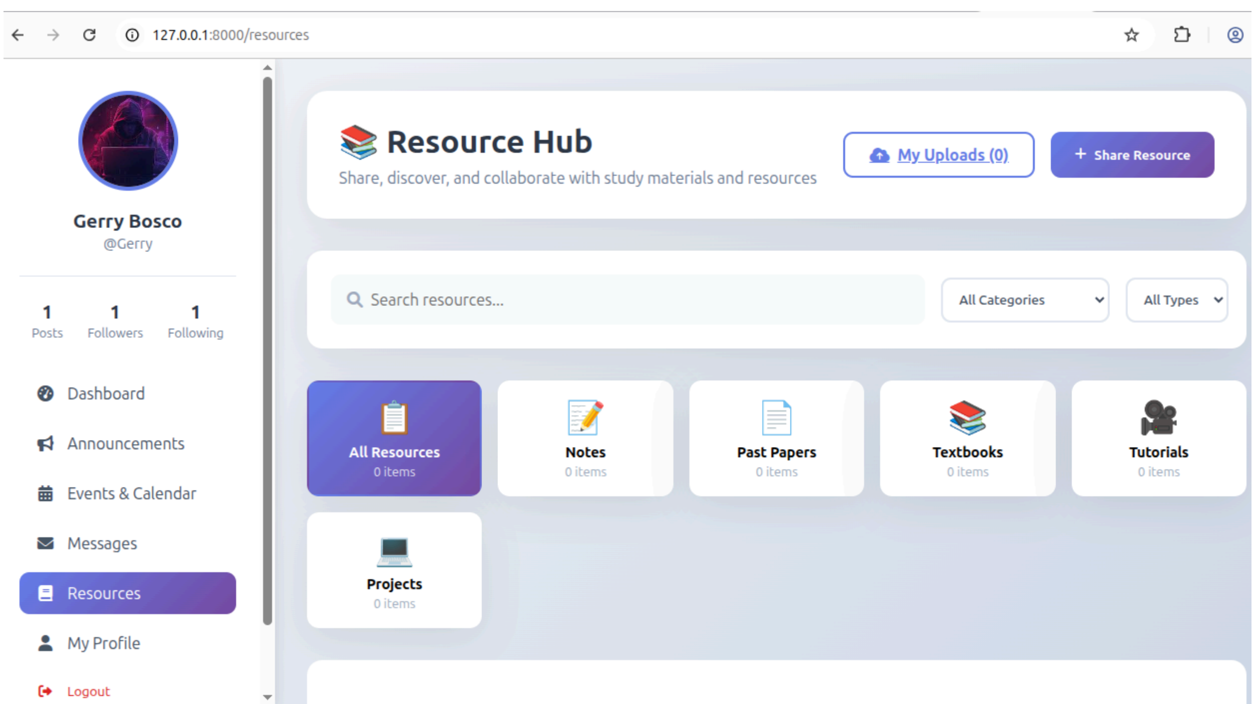
2. Home Feed/ Dashboard



3. Create Post



Sharing a resource



These screenshots provide a visual representation of the system’s main features, showing that the application allows students to log in, create posts, upload resources, and comment, fulfilling its intended purpose of centralized communication and academic resource sharing.